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Unit 1 Exploring MODI Hardware

PDF Documents

PDF Document 1: 1731841746824-Chapter 2_ unit1 Exploring MODI Hardware.pdf

This PDF document is included in the unit materials.

YouTube Links

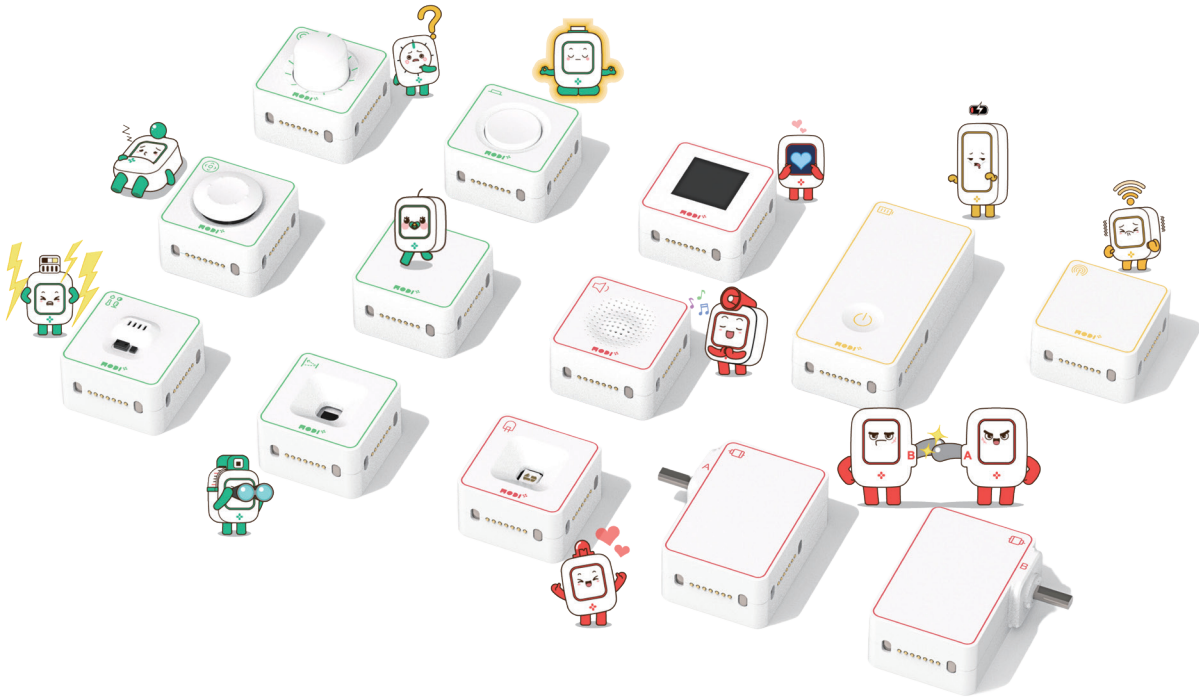
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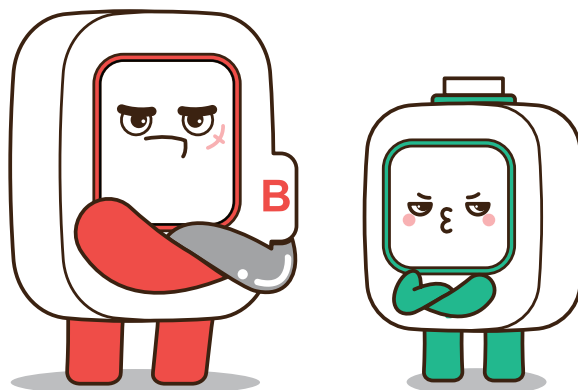
UNIT 1 EXPLORING MODI HARDWARE

Welcome to the World of MODI!

In this unit, we're going to dive into the exciting components that make up the MODI Master Kit. Think of this kit as your toolbox for building and programming your very own robotic creations. Just like a painter needs brushes and paints, a MODI inventor needs modules and ideas!



This exploration into MODI is not merely an introduction to robotics and programming; it's a comprehensive journey into the world of STEAM education. Through engaging with these compact modules, we will gain insights into the foundational concepts that underpin modern technology, enhancing our understanding of how various elements in science, technology, engineering, arts, and mathematics interconnect and apply to real-world devices.





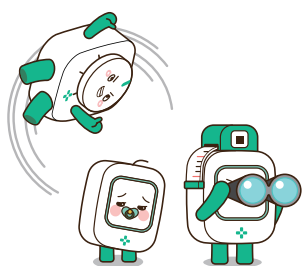
Discovering the MODI Master Kit

Welcome to the exciting world of MODI! The MODI Master Kit is your toolkit for diving into robotics and programming. When you open your MODI Master Kit, you'll find a bunch of colorful modules. Each one has a special job, and they're designed to be easy and fun to use. These modules are the building blocks of any gadget or robot you want to create.

Understanding MODI Modules by Color (3 Types of Modules)

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GREEN: THE COLOR OF INPUT MODULES



These modules are like the device's senses—they're always on the lookout, gathering information from the world. They can sense all sorts of things, such as how loud a sound is, how warm or cold it is, how zzzwet the air is, how far away something is, and much more, just like our senses help us understand the world!

INPUT MODULES



Button

Senses the status of the button



Dial

Senses the rotation of the dial



Environment

Senses temperature, humidity, illuminance, and sound



IMU

Senses slopes (Angles)



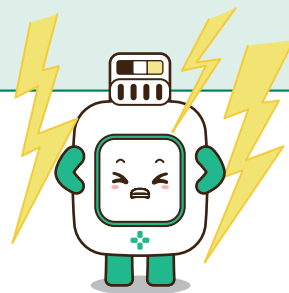
ToF

Senses distance

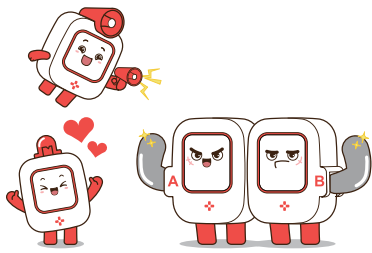


Joystick

Senses the direction of the stick



RED: THE COLOR OF ACTION WITH OUTPUT MODULES



Red MODI modules are the doers—they take information and make something happen. In robotics, we often call these "actuators" because they put things into action. After the green modules pick up information from the environment, like noticing it's gotten dark, the red modules swing into action. They can light up, make sounds, move parts, or even display messages.

OUTPUT MODULES



Speaker

Make sounds



Motor A, B

Rotates



Display

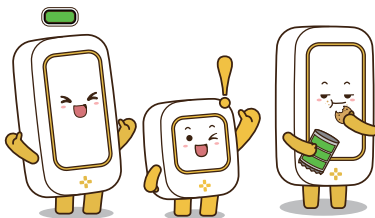
Shows letters and images



LED

Emits light

YELLOW: THE FOUNDATION WITH SETUP MODULES



Yellow MODI modules might be fewer in number, but their role is crucial. They're the backbone of any MODI creation, providing the necessary power and connections to make sure all the modules can work together smoothly. Think of them as the unsung heroes, ensuring that the green and red modules can do their jobs without a hitch. Without these yellow modules, our colorful garden of MODI modules wouldn't be able to come to life.

SETUP MODULES



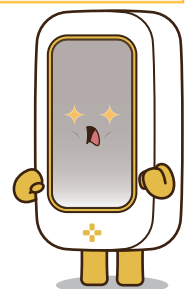
Battery

Supplies energy to the modules



Network

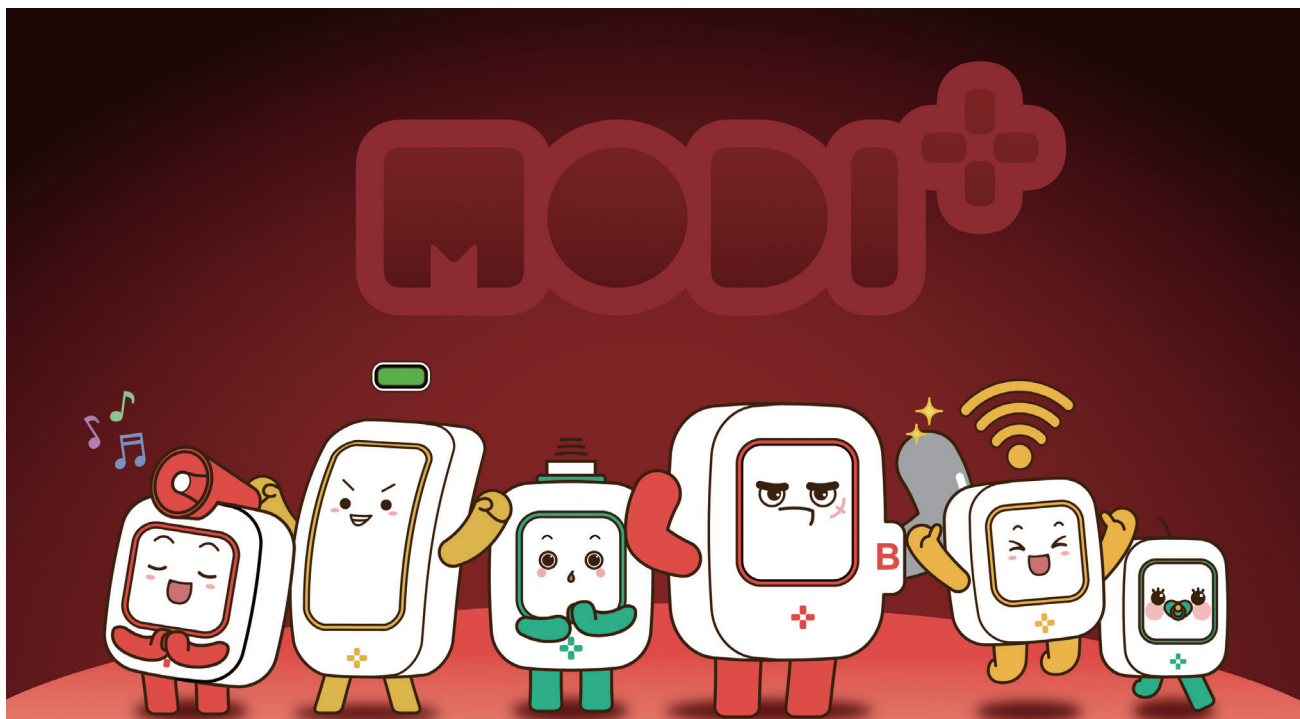
Connects modules to other modules, PCs, tablets, and phones





Bringing It All Together

Think about your smartphone. Have you ever wondered about the many different parts inside it, each doing its own job? From the screen that displays images and videos to the tiny sensors that detect light and movement, every component plays a crucial role. Just like in our MODI projects, it's this teamwork among various components that makes your smartphone so smart and helpful.



Just like we discovered how different parts in a computing system work together, MODI modules do the same thing. Each module has its special job, and when they all work in harmony, amazing things happen. It's important to remember that no single part can do it all on its own.

